

Will Kilgore

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Education

McGill University

Bachelor of Engineering in Mechanical Engineering - GPA: 3.84/4.0

Expected Dec 2027

Montreal, QC

Work Experience

Automated Tire, Inc.

May 2026 – Aug 2026

Mechanical Engineering Intern – Supporting development of an autonomous tire-changing robotic platform

Woburn, MA

- Created technical drawings and assembly documentation in Onshape for 100+ component prototype and production assemblies, reducing technician build time and enabling builds without engineer oversight
- Designed, fabricated, and iterated process-specific fixtures and tooling to resolve assembly issues, reducing production time by 40%
- Optimized and documented changes in electromechanical routing by re-terminating and shortening cable harnesses, improving serviceability across multiple robotic platforms
- Identified and documented 5+ design discrepancies during robotic system assembly, providing structured redline feedback that resolved recurring build issues across 2 design iterations
- Maintained three multi-material FDM printers including calibration, hardware servicing, and failure diagnosis, reducing print failures and achieving same-day prototype turnaround for engineering team
- Implemented a structured inventory system for 1,000+ components, improving traceability and reducing BOM lookup time by up to 10 minutes

Parlee Composites

June 2025 – Present

Manufacturing and Engineering Intern – Improving production of custom high-end carbon fiber bicycles

Beverly, MA

- Conducted vibrational analysis study using analytical models and accelerometer data to evaluate vertical and lateral bicycle frame compliance, informing design changes that improved rider comfort in new frame development
- Built a custom graphical comparison system overlaying generated fit coordinates with 200 existing rider fit data points to enable data-driven component design decisions, reduce required new production part count by 25%, and maintain broad fit coverage
- Developed an Excel-based fit analysis tool generating 150+ stem stack/reach coordinate combinations across varying stem lengths, stem angles, and frame sizes to reduce dealer consultation time
- Introduced new prepreg composite layups that reduced void content and post-processing time, increasing small-part production throughput by up to 30%
- Managed quality control for overseas frame production using inspection checklists and defect tracking, reducing recurring supplier issues through improved defect identification and feedback loops

Projects

McGill Rocket Team

Sept 2023 – Present

Airframe Composites Project Co-Lead

- Co-led composites manufacturing for the aerostructures subteam, coordinating layups, tooling, and assembly integration across a 10-member subteam to deliver flight-ready structures for Launch Canada
- Designed and fabricated molds for a 42" composite nose cone as part of the aerostructures subteam, enabling the transition to a fully student-built 8" rocket platform
- Executed vacuum-assisted resin infusion (VARI) and prepreg layups to manufacture lightweight, high-strength structural body tubes for competition rockets

Carbon Fiber Bike Components

May 2025 – Aug 2025

- Designed and developed custom carbon fiber bicycle components using forged carbon fiber and custom molds to meet specific geometry and weight constraints
- Used SolidWorks, Cura, and 3D printing for rapid prototyping to validate component geometry
- Performed FEA in Abaqus and iterated designs based on simulation and physical fit validation to optimize strength-to-weight performance, reducing weight by 40% from existing component

Skills

Technical Skills: Rapid Prototyping (3D Printing), Design for Manufacturing & Assembly (DFMA), GD&T, FEA, Vibrational Analysis, Experimental Design, Composite Manufacturing (Prepreg, VARI), Cable Routing & Harnessing

Software: SolidWorks, Onshape, Abaqus (FEA), Python (Data Analysis, Visualization), Codex, Excel